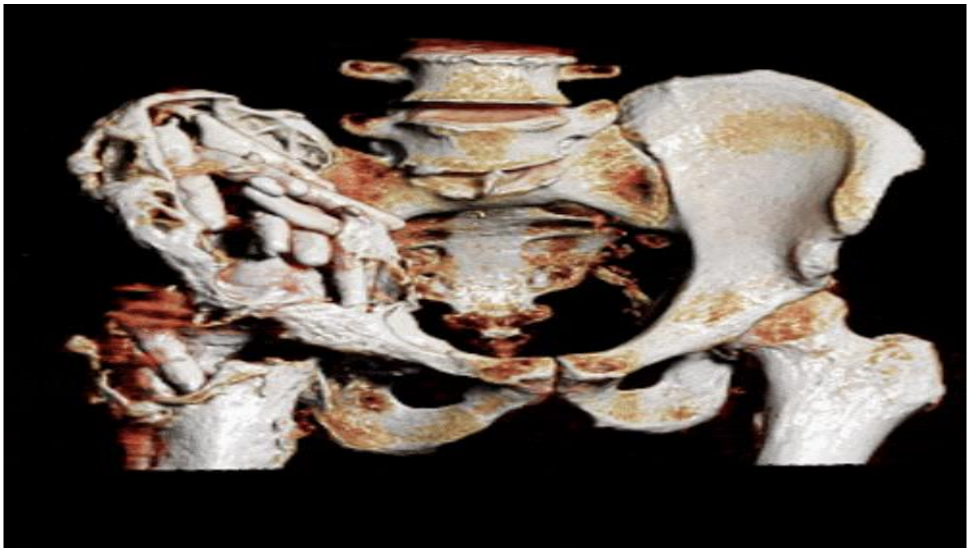
# UNIT I Image Formation& 3-D Imaging

## 1.1 Introduction:

### 1.1.1 What is 3D Imaging?

3D Imaging in healthcare is a collection of techniques and tools used to post-process medical scan data. Unlike 2D imaging, which offers flat, linear views limited to length and width, 3D imaging adds depth, providing a comprehensive representation in three dimensions: length, width, and height (x, y, z coordinates). This depth enables detailed anatomical visualizations of the human body, using technologies such as CT, MRI, and Ultrasound. While 2D methods like X-rays and standard ultrasounds are effective for general imaging, they lack the capability to offer the thorough perspective that 3D imaging provides. Since its introduction in the 1990s it has revolutionized medical diagnosis and treatment, giving healthcare professionals deeper insights into complex health conditions.



### 1.1.2 How is 3D Imaging Created?

The creation of 3D imaging in healthcare is a sophisticated process that hinges on several elements. It begins with scanning a patient's anatomy based on clinical concern. The quality of this scan data is crucial; it follows the "garbage in, garbage out" principle, meaning that the accuracy and resolution of the initial scan data directly impacts the quality of the 3D imaging (1)(2). Key elements like high resolution, contrast enhancement, and the minimization of artifacts significantly influence the overall quality of both the imaging scan and 3D. These aspects should be evaluated before performing the scan to ensure the resulting 3D image is as precise and informative as possible.

The completed scan is loaded into FDA-approved medical imaging software by a radiologist or radiologic technologist, who can use it to produce a variety of 2D and 3D images. This software generally requires at minimum some manual guidance, and it's becoming more common to see automations such as artificial intelligence to assist with reconstruction. The role of radiologists is crucial as they not only oversee the imaging process but also interpret the results.

## 1.2 World and camera coordinates:

### 1.2.1 World Coordinate System

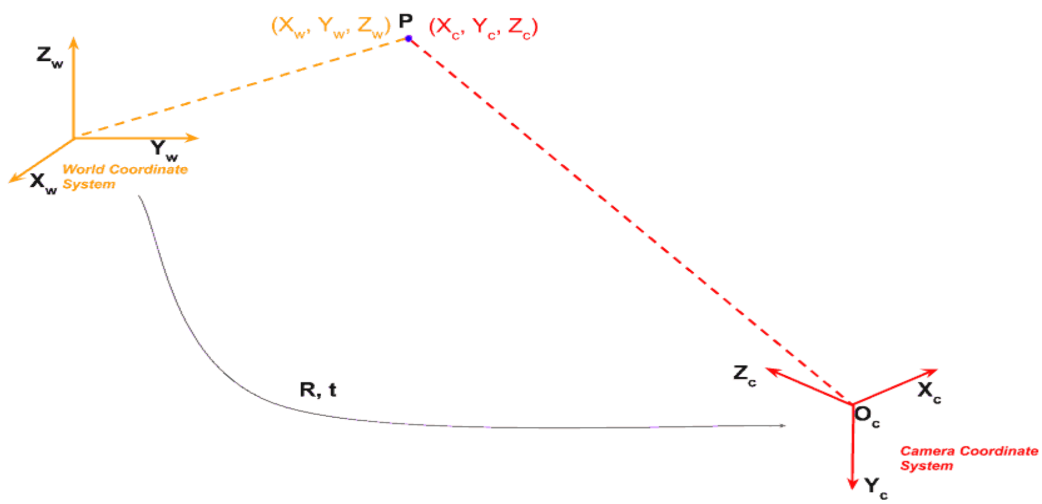



Figure 1: The World Coordinate System and the Camera Coordinate System are related by a Rotation and a translation. These six parameters (3 for rotation, and 3 for translation ) are called the extrinsic parameters of a camera.

To define locations of points in the room we need to first define a coordinate system for this room. It requires two things

1. Origin : We can arbitrarily fix a corner of the room as the origin $(0, 0, 0)$.
2. X, Y, Z axes : We can also define the X and Y axis of the room along the two dimensions on the floor and the Z axis along the vertical wall.

Using the above, we can find the 3D coordinates of any point in this room by measuring its distance from the origin along the X, Y, and Z axes.

This coordinate system attached to the room is referred to as the World Coordinate System. In Figure 1, it is shown using orange colored axes. We will use bold font ( e.g. $\mathbf{X_W}$ ) to show the axis, and regular font to show a coordinate of the point ( e.g. $X_w$ ).

Let us consider a point P in this room. In the world coordinate system, the coordinates of P are given by $(X_w, Y_w, Z_w)$. You can find $X_w$, $Y_w$, and $Z_w$ coordinates of this point by simply measuring the distance of this point from the origin along the three axes.

### 1.2.2. Camera Coordinate System

Now, let's put a camera in this room.

The image of the room will be captured using this camera, and therefore, we are interested in a 3D coordinate system attached to this camera.

If we had put the camera at origin of the room, and align it such that its X, Y, and Z axes aligned with the $\mathbf{X_W}$, $\mathbf{Y_W}$, and $\mathbf{Z_W}$ axes of the room, the two coordinate systems would be the same.

However, that is an absurd restriction. We would want to put the camera anywhere in the room and it should be able to look anywhere. In such a case, we need to find the relationship between the 3D room (i.e. world) coordinates and the 3D camera coordinates.

Let's say our camera is located at some arbitrary location $(t_X, t_Y, t_Z)$ in the room. In technical jargon, we can the camera coordinate is translated by $(t_X, t_Y, t_Z)$ with respect to the world coordinates.

The camera may be also looking in some arbitrary direction. In other words, we can say the camera is rotated with respect to the world coordinate system.

Rotation in 3D is captured using three parameters —- you can think of the three parameters as yaw, pitch, and roll. You can also think of it as an axis in 3D ( two parameters ) and an angular rotation about that axis (one parameter).

However, it is often convenient for mathematical manipulation to encode rotation as a 3×3 matrix. Now, you may be thinking that a 3×3 matrix has 9 elements and therefore 9 parameters but rotation has only 3 parameters. That's true, and that is exactly why any arbitrary 3×3 matrix is not a rotation matrix. Without going into the details, let us for now just know that a rotation matrix has only three degrees of freedom even though it has 9 elements.

Back to our original problem. The world coordinate and the camera coordinates are related by a rotation matrix $\mathbf{R}$ and a 3 element translation vector $\mathbf{t}$

What does that mean?

It means that point P which had coordinate values $(X_w, Y_w, Z_w)$ in the world coordinates will have different coordinate values $(X_c, Y_c, Z_c)$ in the camera coordinate system. We are representing the camera coordinate system using red color.

The two coordinate values are related by the following equation.

(1) $$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \mathbf{R} \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + \mathbf{t}$$

Notice that representing rotation as a matrix allowed us to do rotation with a simple matrix multiplication instead of tedious symbol manipulation required in other representations like yaw, pitch, roll. I hope this helps you appreciate why we represent rotations as a matrix.

Sometimes the expression above is written in a more compact form. The 3×1 translation vector is appended as a column at the end of the 3×3 rotation matrix to obtain a 3×4 matrix called the Extrinsic Matrix.

(2) $$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \left[\mathbf{R} | \mathbf{t}\right] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

where, the extrinsic matrix $\mathbf{P}$ is given by

(3) $$\mathbf{P} = \left[\mathbf{R} | \mathbf{t}\right]$$

Homogeneous coordinates : In projective geometry, we often work with a funny representation of coordinates where an extra dimension is appended to the coordinates. A 3D point $(X, Y, Z)$ in cartesian coordinates can written as $(X, Y, Z, 1)$ in homogenous coordinates. More generally, a point in homogenous coordinate $(X, Y, Z, W)$ is the same as the point $(X/W, Y/W, Z/W)$ in cartesian coordinates. Homogenous

COORDINATES ALLOW US TO REPRESENT INFINITE QUANTITIES USING FINITE NUMBERS. FOR EXAMPLE, THE POINT AT INFINITY CAN BE REPRESENTED AS $(1, 1, 1, 0)$ IN HOMOGENOUS COORDINATES. YOU MAY NOTICE THAT WE HAVE USED HOMOGENOUS COORDINATES IN EQUATION 2 TO REPRESENT THE WORLD COORDINATES.

## 1.3 IDEAL IMAGING: PERSPECTIVE PROJECTION

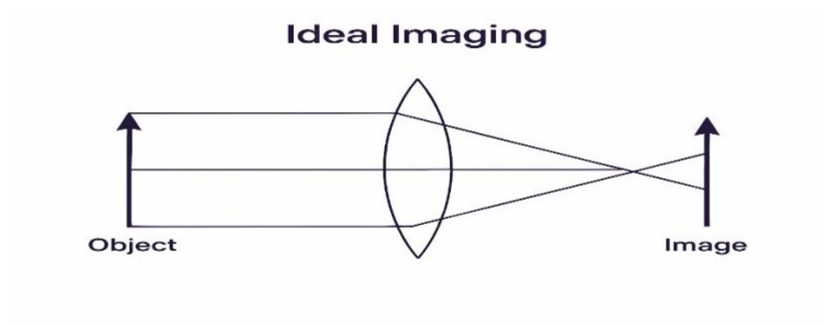
### IDEAL IMAGING:

IDEAL IMAGING IS A THEORETICAL CONCEPT IN OPTICS AND IMAGING SYSTEMS WHERE THE IMAGE FORMATION PROCESS IS ASSUMED TO BE PERFECT WITHOUT ANY PHYSICAL LIMITATIONS OR DISTORTIONS.

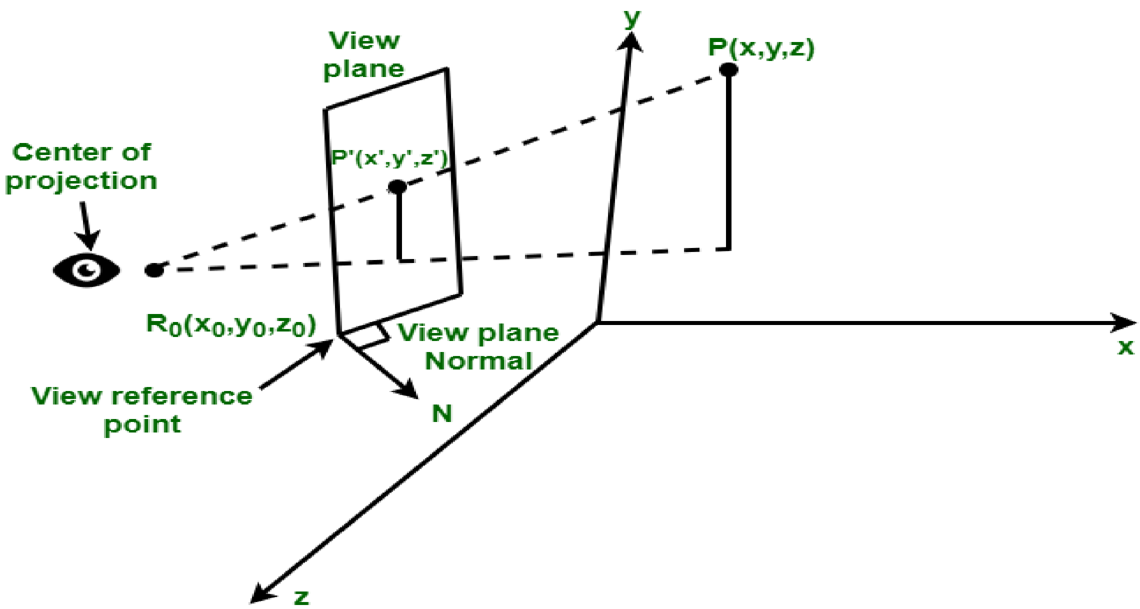
### DEFINITION:

IDEAL IMAGING REFERS TO A SITUATION WHERE AN OPTICAL SYSTEM (LIKE A LENS OR MIRROR) FORMS A PERFECT IMAGE OF AN OBJECT — ONE THAT EXACTLY REPLICATES ALL THE DETAILS OF THE OBJECT WITH:

- INFINITE SHARPNESS
- NO DISTORTION
- PERFECT BRIGHTNESS AND CONTRAST
- NO NOISE OR ABERRATIONS




IN PERSPECTIVE PROJECTION THE CENTER OF PROJECTION IS AT FINITE DISTANCE FROM PROJECTION PLANE. THIS PROJECTION PRODUCES REALISTIC VIEWS BUT DOES NOT PRESERVE RELATIVE PROPORTIONS OF AN OBJECT DIMENSIONS. PROJECTIONS OF DISTANT OBJECT ARE SMALLER THAN PROJECTIONS OF OBJECTS OF SAME SIZE THAT ARE CLOSER TO PROJECTION PLANE. THE PERSPECTIVE PROJECTION CAN BE EASILY DESCRIBED BY THE FOLLOWING FIGURE:

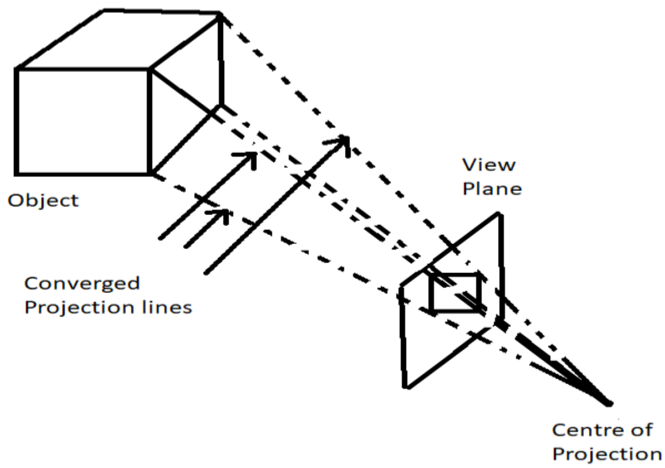



1. **Center of Projection** - It is a point where lines or projection that are not parallel to projection plane appear to meet.
2. **View Plane or Projection Plane** - The view plane is determined by :
   - View reference point $R_0(x_0, y_0, z_0)$
   - View plane normal.
3. **Location of an Object** - It is specified by a point P that is located in world coordinates at (x, y, z) location. The objective of perspective projection is to determine the image point P' whose coordinates are (x', y', z')

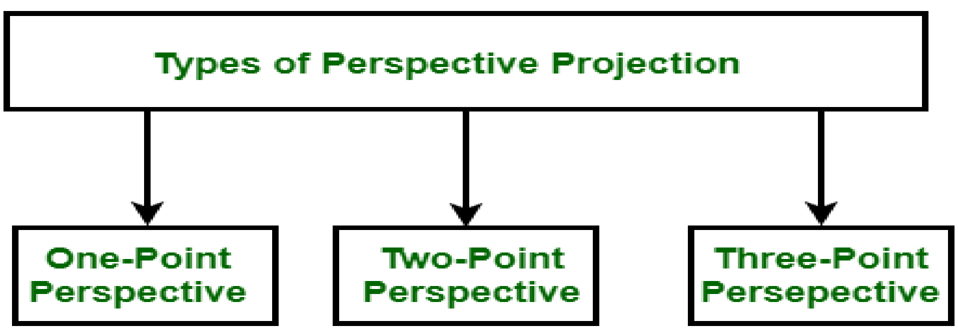
The perspective projection, on the other hand, produces realistic views but does not preserve relative proportions.

In perspective projection, the lines of projection are not parallel. Instead, they all converge at a single point called the center of projection or projection reference point.

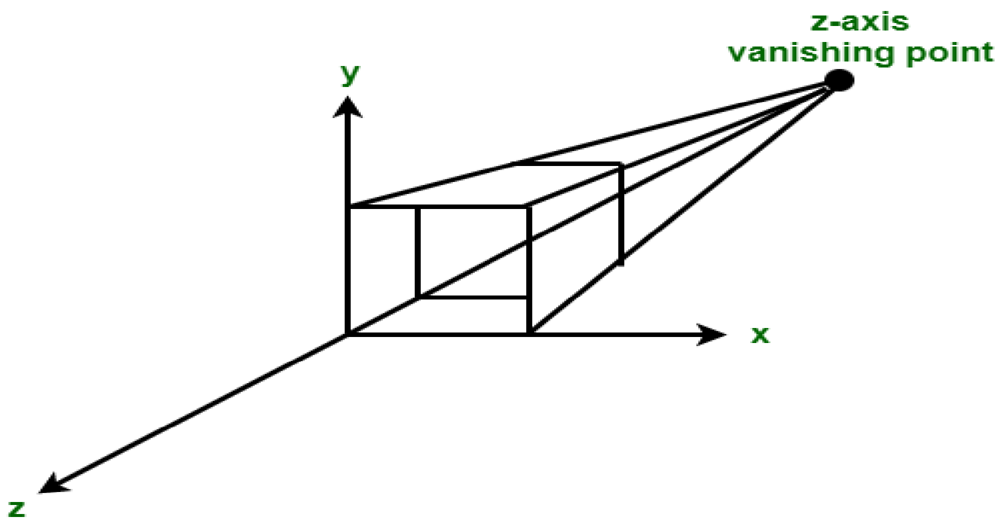
The object positions are transformed to the view plane along these converged projection lines and the projected view of an object is determined by calculating the intersection of the converged projection lines with the view plane, as shown below figure




**Types of Perspective Projection:** Classification of perspective projection is on basis of vanishing points (It is a point in image where a parallel line through center of projection intersects view plane.). We can say that a vanishing point is a point where projection line intersects view plane. The classification is as follows:

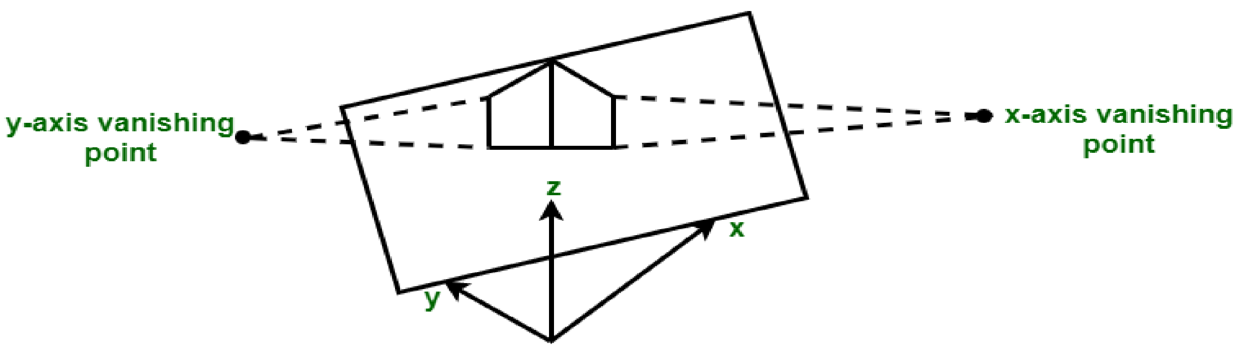



- **One Point Perspective Projection** - One point perspective projection occurs when any of principal axes intersects with projection plane or we can say when projection plane is perpendicular to principal axis.




In the above figure, z axis intersects projection plane whereas x and y axis remain parallel to projection plane.

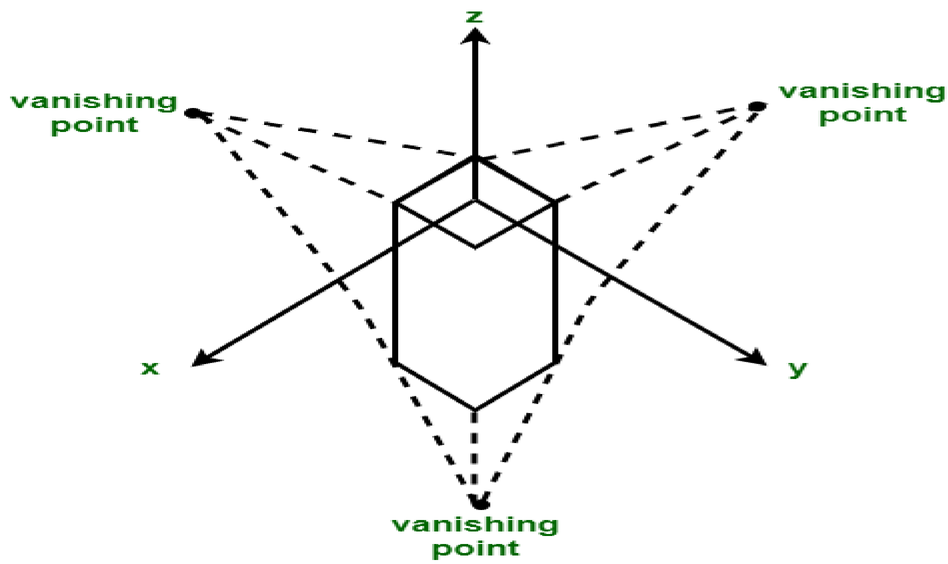
- **Two Point Perspective Projection** - Two point perspective projection occurs when projection plane intersects two of principal axis.




In the above figure, projection plane intersects x and y axis whereas z axis remains parallel to projection plane.

- **Three Point Perspective Projection** - Three point perspective projection occurs when all three axis intersects with projection plane. There is no

ANY PRINCIPAL AXIS WHICH IS PARALLEL TO PROJECTION PLANE.




## 1.4 Real Imaging

Real Imaging refers to the actual process of image formation using real-world optical systems like cameras, microscopes, telescopes, or the human eye, where imperfections and limitations affect the final image quality.

Definition:

Real imaging occurs when physical devices (lenses, mirrors, sensors) form an image of an object, but the image is imperfect due to:

- Optical limitations
- Physical interference
- Environmental factors

Key Characteristics of Real Imaging:

| Feature | Description |
|---|---|
| Lens Aberrations | Images may be blurred or distorted due to spherical or chromatic aberrations. |
| Diffraction Effects | Light behaves as a wave, limiting the sharpness due to diffraction. |

| Feature | Description |
|---|---|
| Sensor Noise | Electronic sensors (like CCD or CMOS) introduce noise into the image. |
| Finite Resolution | Resolution is limited by pixel size, lens quality, and light conditions. |
| Distortion | Lines may appear bent or curved due to imperfect lens design (e.g., barrel distortion). |
| Color Inaccuracy | Light absorption, scattering, or poor sensors can alter true colors. |
| Depth of Field | Only parts of the image are in sharp focus at a time. |
| Light Scattering | Real-world imaging is affected by dust, fog, and atmospheric conditions. |

Examples:

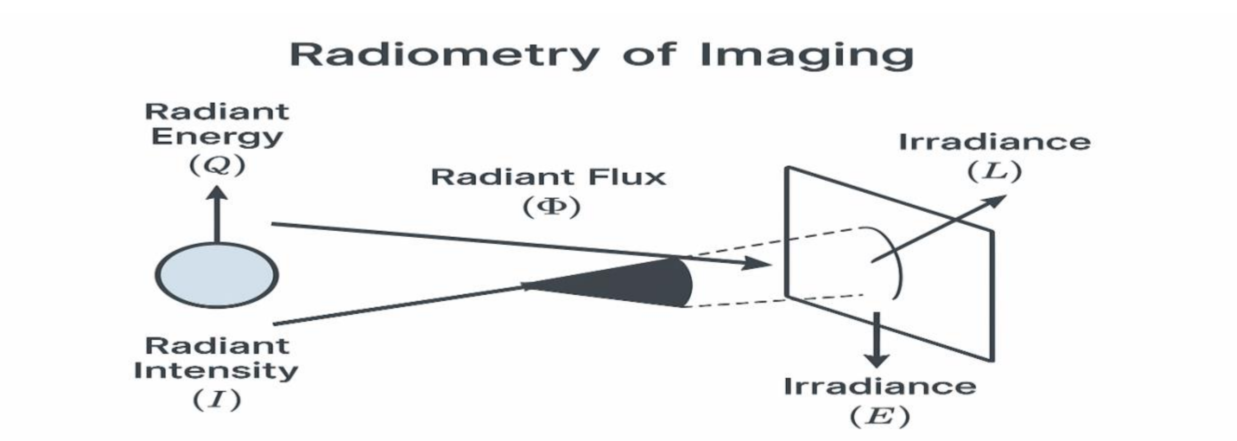
- A photo taken by a mobile phone camera where edges are slightly blurred and colors slightly off.
- An image seen through a microscope, affected by lens quality and sample preparation.
- CCTV footage with grainy images in low light due to sensor limitations.

## 1.5 Radiometry of Imaging

**Radiometry of Imaging** is a key concept in image science, computer vision, and optics. It refers to the *measurement of electromagnetic radiation*, including visible light, in terms of its energy content and distribution. Radiometry quantifies how much light energy is captured in an imaging system, without concern for human perception (unlike photometry, which is based on human vision).

**Radiometric Quantities**

| Quantity | Symbol | Unit | Description |
|---|---|---|---|
| **Radiant Energy** | QQQ | joule (J) | Total energy emitted, transferred, or received. |
| **Radiant Flux / Power** | Φ\PhiΦ | watt (W = J/s) | Rate of energy flow per second. |
| **Radiant Intensity** | III | W/sr | Power emitted per unit solid angle (used for point sources). |
| **Irradiance** | EEE | W/m² | Power received per unit area on a surface. |
| **Radiance** | LLL | W/(m²·sr) | Power per unit projected area per unit solid angle — **most important in imaging**. |




## 1.5 Liner System Theory of Imaging

The **Linear System Theory of Imaging** models the image formation process as a **linear and shift-invariant system**, which is widely used in optics, image processing, and computer vision.

**1.Linearity**

- The system obeys the **superposition principle**:

H{a·f1(x)+b·f2(x)}=a·H{f1(x)}+b·H{f2(x)}H\{a \cdot f_1(x) + b \cdot f_2(x)\} = a \cdot H\{f_1(x)\} + b \cdot H\{f_2(x)\}H{a·f1(x)+b·f2(x)}=a·H{f1(x)}+b·H{f2(x)}

where HHH is the imaging system, and f(x)f(x)f(x) is the input signal.

2. **Shift-Invariance**
   - The system's behavior does **not change with spatial shifts**.

H{f(x−x0)}=[H{f(x)}](x−x0)H\{f(x - x_0)\} = [H\{f(x)\}](x - x_0)H{f(x−x0)}=[H{f(x)}](x−x0)

3. **Impulse Response / Point Spread Function (PSF)**
   - The response of the system to a **point source** (ideal impulse).
   - Denoted as h(x)h(x)h(x), it characterizes the system completely under LSI conditions.
4. **Convolution Operation**
   - The output image is the **convolution** of the input object with the system’s PSF:

g(x)=f(x)∗h(x)g(x) = f(x) * h(x)g(x)=f(x)∗h(x)

5. **Frequency Domain (Fourier Transform)**
   - Convolution in space domain = Multiplication in frequency domain:

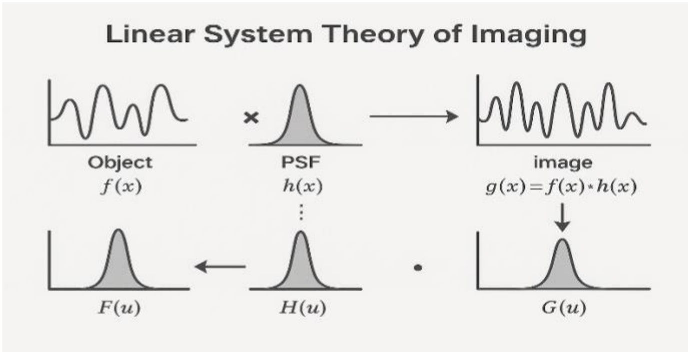
G(u)=F(u)·H(u)G(u) = F(u) \cdot H(u)G(u)=F(u)·H(u)

- H(u)H(u)H(u): **Optical Transfer Function (OTF)** — the Fourier transform of the PSF.

**Diagram Elements to Include**

- Object f(x)f(x)f(x)
- PSF h(x)h(x)h(x)
- Image g(x)=f(x)∗h(x)g(x) = f(x) * h(x)g(x)=f(x)∗h(x)

- Fourier domain: G(u)=F(u)·H(u)G(u) = F(u) \cdot H(u)G(u)=F(u )·H(u)




## 1.6 Homogeneous Coordinates

**Homogeneous coordinates** are an extension of regular Cartesian coordinates used in projective geometry, computer graphics, and computer vision. They simplify the mathematics of transformations like **translation, rotation, scaling**, and **perspective projection**.

### Definition

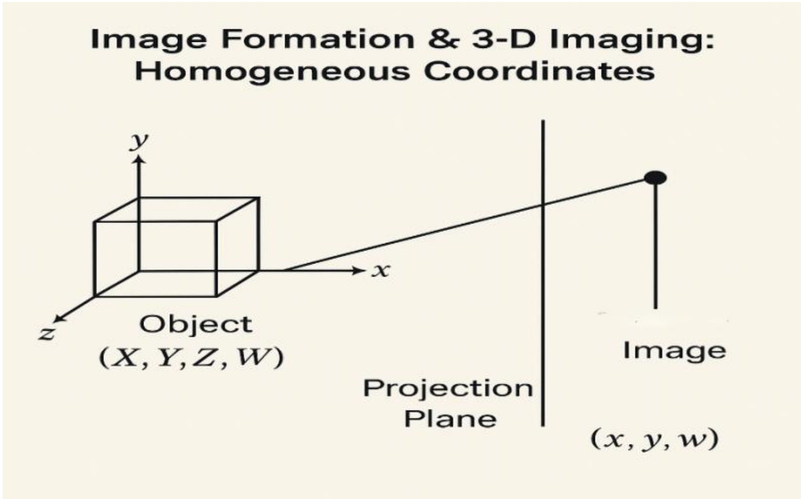
A point in **2D Cartesian space** (x,y)(x, y)(x,y) is represented in **homogeneous coordinates** as:

(x,y)→(x,y,1)(x, y) \rightarrow (x, y, 1)(x,y)→(x,y,1)

In general:

- A point (x,y)(x, y)(x,y) becomes (wx,wy,w)(wx, wy, w)(wx,wy,w)
- A point (x,y,z)(x, y, z)(x,y,z) becomes (wx,wy,wz,w)(wx, wy, wz, w)(wx,wy,wz,w)
- To convert back: divide by www:

  (X,Y,W)→(XW,YW)(X, Y, W) \rightarrow \left(\frac{X}{W}, \frac{Y}{W}\right)(X,Y,W)→(WX,WY)




## 1.6. Introduction to 3-D Imaging: Basics

**3-D Imaging** refers to capturing, processing, and visualizing real-world objects in three dimensions. Unlike 2D imaging (which captures only height and width), 3D imaging also includes **depth**, offering a more realistic and spatially accurate representation of the scene or object.

**1. Depth (Z-axis)**

- Adds a third dimension to traditional 2D images.
- Represents the distance of objects from the viewpoint or camera.

**2. Coordinate Systems**

- **3D point**: (X,Y,Z)(X, Y, Z)(X,Y,Z)
- Often expressed in **homogeneous coordinates**: (X,Y,Z,1)(X, Y, Z, 1)(X,Y,Z,1)
- Used to transform, project, and map objects from 3D world to 2D image plane.

**3. Projection**

- Converts 3D points into 2D image coordinates:

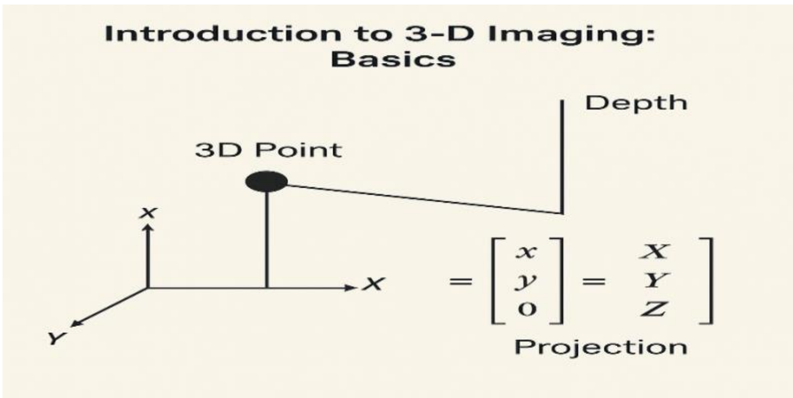
  [xyw]=P[XYZ1]\begin{bmatrix} x \\ y \\ w \end{bmatrix} = \mathbf{P} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}xyw=PXYZ1

  Where P\mathbf{P}P is the **camera projection matrix**.

### Applications:

- **Medical Imaging**: CT, MRI, and 3D ultrasound

- **Robotics & Drones**: Environment mapping and navigation
- **AR/VR & Gaming**: Realistic object and world modeling
- **Industrial Inspection**: Precision measurement and defect detection
- **Autonomous Vehicles**: LiDAR and stereo for obstacle detection




## 1.7 Depth from Triangulation

**Triangulation** is a geometric technique used to determine the **depth (distance)** of a point in 3D space by **observing it from multiple viewpoints**.

- It works like human eyes: each eye sees an object from a slightly different angle.
- In imaging, **two cameras** placed side by side can do the same thing: this is called **stereo vision**.

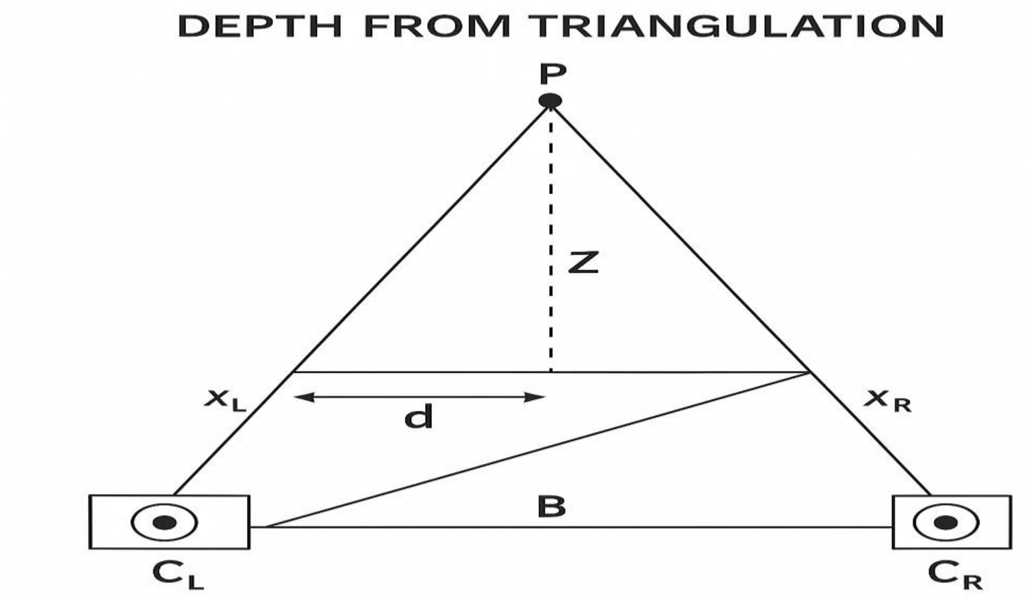
**Components:**

- **Two cameras** (Left & Right)
- **Common focal length** fff
- **Known baseline** BBB: the distance between the optical centers of the two cameras
- A **3D point** PPP observed in both images at:
  - xLx_LxL: x-coordinate in **left image**
  - xRx_RxR: x-coordinate in **right image**
- From both camera views, we get the **disparity**:
- Disparity (d)=xL−xR\text{Disparity } (d) = x_L - x_RDisparity (d)=xL−xR
- Now, using similar triangles, we derive the **depth formula**:
- Z=f·BdZ = \frac{f \cdot B}{d}Z=df·B

| Term | Symbol | Unit | Meaning |
|---|---|---|---|
| Focal length | ff | mm or px | Distance from camera center to sensor |
| Baseline | BB | mm | Distance between the two cameras |
| Disparity | dd | px | Difference in x-coordinates |
| Depth | ZZ | mm | Distance of point from the camera |

## Step-by-Step Working

1. **Capture Images from Left and Right cameras.**
2. **Detect Matching Points in both images (feature matching).**
3. **Calculate Disparity d=xL−xRd = x_L - x_Rd=xL−xR**
4. **Apply Depth Formula Z=f·BdZ = \frac{f \cdot B}{d}Z=df·B**
5. **Get the 3D coordinates of the point.**




## 1.8. Depth from Time-of-Flight

**What is Time-of-Flight (ToF)?**
**Time-of-Flight (ToF) is a technique for measuring distance (depth) to an object by calculating the time it takes for light to travel from a source, bounce off the object, and return to a sensor.**
**It's based on the principle:**
**Distance = Speed × Time**

**This principle is used in physics (like radar and sonar) and has been adapted into optical 3D imaging.**

### 2. How Does It Work?

1. **A light source (usually near-infrared) emits a light pulse or a modulated wave.**
2. **The light hits an object and reflects back.**
3. **A sensor measures how long the light took to return or how much its phase shifted.**
4. **This delay or shift is converted to depth information.**

### 3. ToF Mathematical Model

**➤ Direct ToF: Pulse-Based**

**Z=c·t2Z = \frac{c \cdot t}{2}Z=2c·t**

- **ZZZ: Depth to object**
- **ccc: Speed of light ≈ 3×108 m/s3 \times 10^8 \text{ m/s}3×108 m/s**
- **ttt: Round-trip time of flight**
- **Division by 2 accounts for light going to and back from the object**

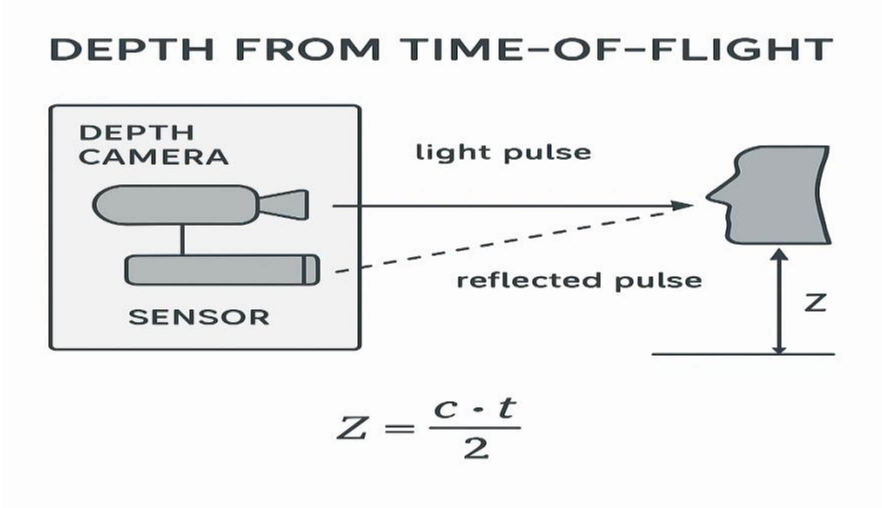
**➤ Indirect ToF: Modulated Light (Phase Shift)**

**Z=c·Δϕ4πfZ = \frac{c \cdot \Delta \phi}{4\pi f}Z=4πfc·Δϕ**

- **Δϕ\Delta \phiΔϕ: Phase shift between emitted and received light**
- **fff: Modulation frequency of the light wave**

### 4. Types of ToF Systems

| Type | How it works | Pros | Cons |
|---|---|---|---|
| Direct ToF | Measures actual time delay of reflected pulse | Good for long-range | Needs fast and precise timing |
| Indirect ToF | Measures phase shift of continuous modulated wave | Power-efficient | Limited range and resolution |




## 1.9 Depth from Phase: Interferometry

**Interferometry is a technique that uses the interference of waves (typically light, but also sound or radio) to measure very small displacements, surface irregularities, or changes in optical path lengths with extreme precision.**

**When applied to imaging, interferometry enables 3D surface measurement by evaluating the phase difference between two or more coherent beams of light.**

**When a light wave reflects off a surface, its phase changes depending on how far it traveled. By comparing the reference beam with the measurement beam reflected off the object, we can compute the phase difference:**

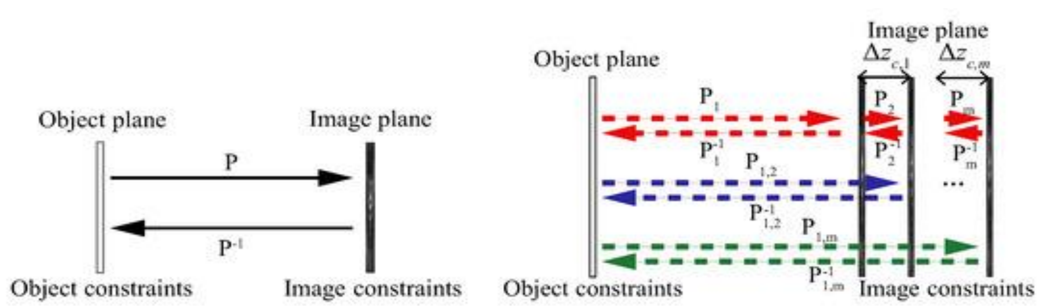
**Δϕ=4πdλ\Delta \phi = \frac{4\pi d}{\lambda}Δϕ=λ4πd**

**Where:**

- **Δϕ\Delta \phiΔϕ = Phase shift**
- **ddd = Optical path difference (i.e., depth)**
- **λ\lambdaλ = Wavelength of the laser**

**The depth is then recovered as:**

**d=λ·Δϕ4πd = \frac{\lambda \cdot \Delta \phi}{4\pi}d=4πλ·Δϕ**




(a) (b)

## 1.10. Shape from Shading

**shape from Shading** is a classic **monocular 3D reconstruction** technique that aims to infer the **3D shape of an object** from **brightness variations** (shading) in a single grayscale image.

- The **brightness** of each pixel is influenced by how the surface is **oriented relative to a light source**.
- SfS works by **inverting this process** — estimating **surface normals** and then integrating them to recover the **surface depth**.

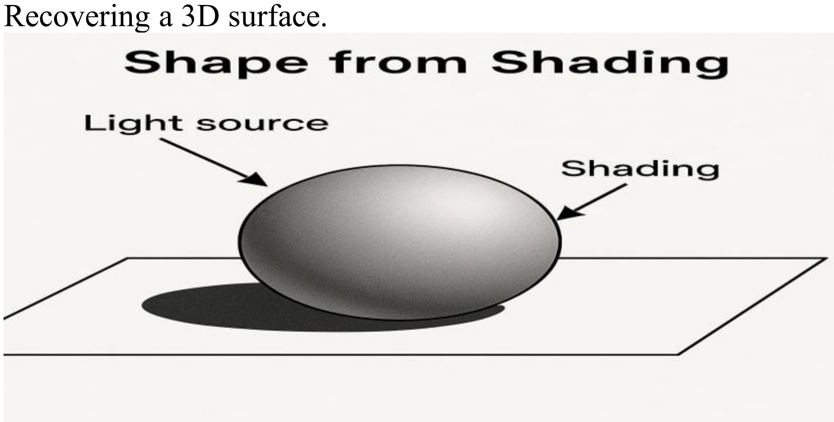
### Principles

- Brightness in an image depends on:
- **Surface orientation** (normal vectors).
- **Light source direction**.
- **Reflectance model** (usually Lambertian).
- **Brightness I(x, y) ≈ ρ (N · S)**
  where:
  ρ = albedo (reflectivity),
  N = surface normal,
  S = light source vector,
  · = dot product.

  **Mathematical Model**

  Given the brightness I(x,y)I(x, y)I(x,y), estimate the **depth z(x,y)z(x, y)z(x,y)** of each pixel by:

1. Estimating surface normals.
2. Integrating gradients ∂z∂x,∂z∂y\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}∂x∂z,∂y∂z.
3. Recovering a 3D surface.



## Depth from Multiple Projections: Tomography

**Tomography** is a technique to recover the **3D internal structure** of an object by combining its **2D projections** taken at multiple angles.

It is widely used in **medical imaging, materials science, and industry**.

**Basic Principle**

- Each **2D projection** is a **line integral** through the object (i.e., it sums up values like density along a path).
- Multiple such projections (from different angles) allow reconstruction of the **original 3D function** representing the internal structure.

**Mathematical Foundation**

➤ **Radon Transform**

The Radon transform converts a 2D function f(x,y)f(x, y)f(x,y) into its projections:

R(θ,s)=∫−∞∞∫−∞∞f(x,y)⋅δ(s−xcos⁡θ−ysin⁡θ) dx dyR(\theta, s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \cdot \delta(s - x\cos\theta - y\sin\theta)\, dx\, dyR(θ,s)=∫−∞∞∫−∞∞f(x,y)⋅δ(s−xcosθ−ysinθ)dxdy

- θ\thetaθ: projection angle
- sss: distance along the projection
- δ\deltaδ: Dirac delta function

**Types of Tomography**

| **Type** | **Description** | **Imaging Medium** |
|---|---|---|
| **CT (Computed Tomography)** | X-ray projections from multiple angles | X-rays |
| **MRI (Magnetic Resonance Imaging)** | Uses magnetic fields and radio waves | RF signal |
| **PET (Positron Emission Tomography)** | Detects gamma rays from radioactive tracers | Gamma rays |
| **SPECT (Single Photon Emission CT)** | Gamma-emitting tracers, like PET | Gamma rays |

| Type | Description | Imaging Medium |
|---|---|---|
| **Optical Tomography** | Light transmission through tissue | Light |
| **Electron Tomography** | Electron beams used in nanostructures | Electrons |

## Workflow of Tomographic Imaging

**Step 1: Acquire Projections**

- The object is scanned at different angles.
- Each projection records the cumulative data along rays.

**Step 2: Build Sinogram**

- A **sinogram** is a 2D plot showing projection data over angles.
- Each horizontal line = one projection.

**Step 3: Reconstruct Image**

- Use algorithms to calculate original slice from projections.

Figure: Tomography